

# Eastern Mediterranean University – Computer Engineering Department

## CMPE/CMSE/AING231 – Data Structures

### Assignment #1

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Miss. Hatice is a primary school teacher and teaches numbers in her classroom. For this reason, every day she asks her students to say 10 numbers in order, starting with a random number. (e.g., 5, 7, 9, 15, 16, 20, 21, 23, 30, 55.) Miss. Hatice's class is very crowded and in order to save time and allow the students to practice on their own in their free time, she needs a program that runs on the computer and checks whether the 10 numbers entered by the students are in order. Develop a program for Miss. Hatice by considering the following restrictions.

- You should implement your solution with a stack.
- It should be a menu-driven program that includes menu items such as “New set of number entry”, “Exit”, etc.
- The number set should be checked after the student enters each number set.
- These numbers should be stored in the stack while the student enters the number set.
- The stack should be used when checking the number set; push and pop functions.
- The set of numbers entered by the student AND whether they are correct or not should be shown on the screen. If it is not correct, the correct set of numbers should be given.
- You must "initialize" your stack with each new set of number entries.
- Since it is a program that primary school students will use, make sure that the texts and messages that the user will see in your program are suitable for them.
- Suppose that the student will enter 10 numbers at a time.
- Ascending order should be used for the numbers.